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Morphometric study of lake basins from the Indian subcontinent: A critical review

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Lakes from the Indian subcontinent are ecosystems of vital importance as they contain high levels of biodiversity which is threatened by multiple anthropogenic pressures and the impacts of ongoing global warming. The present review article examines the morphometric features of 35 lakes from the Indian subcontinent and explores the links between lake morphometry, sediment distribution in the lake bottom, and degradation of the lake ecosystem. For this purpose, a series of lake morphometric parameters such as Schindler's ratio (SR), dynamic ratio (DR), relative depth (RD), shoreline development index (SDI) and energy factor (EF) were calculated. In addition, a principal component analysis (PCA) was performed on morphometric parameters, which identified (i) lake size and (ii) water depth as the key parameters that influence sediment distribution at the lake bottom. Our findings also implied that the lakes with higher SR values (e.g., Shilloi, Anchar, Pulicat, and Tso Kar Lakes) are prone to water quality degradation due to inputs from much larger catchments as compared to their lake water volume. However, a detailed understanding of modern lake processes (such as water mixing and evaporation) is required to decipher the link between the degradation status of the lake and the cumulative response of both the lake and catchment processes. Furthermore, the high fetch length and DR for relatively shallow lakes such as Chilika, Kolleru, Loktak, and Pulicat suggest that the sediment distribution is more likely to be affected by wind-induced mixing effects, whereas in deeper lakes (e.g., Tso Moriri, Phoksundo, Rara), the role of wave action is minimal. Conversely, in the small-shallow lakes, the role of peripheral wave activity is the dominant phenomenon that controls the sediment resuspension and redistribution within the basin. Further, the majority of the studied lakes show high SDI values (>1.5) suggesting irregular shorelines, which favours the development of complex littoral habitats, either due to anthropogenic activity or increased sediment supply from the surface runoff. This parameter also highlighted the importance of an effective sampling design that reflects the spatial variability of sediment behaviour in various parts of the lake. To the best of our knowledge, this is the first attempt in the Indian subcontinent to utilise morphometric indices for understanding the lake's degradation status and the role of excessive anthropogenic activity in influencing shoreline habitats.

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1. Introduction

Lake systems are ideal natural archives for understanding the modern and past climatic conditions of the wider region (Mishra *et al.* 2014; Opitz *et al.* 2015; Xiao *et al.* 2015; Ankit *et al.* 2017; Ratnayake *et al.* 2017). The lacustrine basins are exposed to several external (wind, atmospheric temperature, solar irradiance) as well as internal (stream inflow, water volume, and physico-chemical characteristics) factors that regulate key limnological processes and local hydrological balance (Hutter *et al.* 2010). Additionally, the geomorphological and the morphometric parameters along with the catchment activity (e.g., weathering, erosion) are known to play a major role in controlling the lake's internal processes (e.g., mixing and physico-chemical condition), trophic status, hydrogeochemistry and sediment characteristics in the basin (Kalff 2002; Håkanson 2005; Nõges 2009; Stefanidis and Papastergiadou 2012).

From the Indian subcontinent, in lacustrine settings, several previous studies have focussed on the calibration of climate-sensitive proxies according to changes in modern environmental conditions, which is being used to reconstruct past climate variability (Basavaiah *et al.* 2014; Mishra *et al.* 2014; Sandeep *et al.* 2017; Quamar 2019). These studies have largely focussed on the factors governing the sediment supply into the lake basin in response to changes in the natural as well as anthropogenic factors (Mishra *et al.* 2014; Sandeep *et al.* 2017; Quamar 2019; Prasad *et al.* 2020). However, little attention has been devoted to investigating the role of lake basin morphometry in influencing physical (sediment supply) and biological (macrophytes growth) activity in the lake basins (Lawniczak-Malińska *et al.* 2018). Another issue while working with these lakes is the unavailability or scarcity of modern/morphometric data. This leads to speculation or misinterpretation of proxy behaviour according to changes in the lake settings. Therefore, before conducting paleoclimate research, baseline information between the lake basin morphometry and its role in the limnological processes is much needed for further studies (Graf 1982; Vincent 2009).

In this review work, we are focussing on the morphometric characteristics of 35 lake basins from the Indian subcontinent to (i) investigate the role of lake basin morphometry and catchment relief in controlling the sediment distribution within the lake basin; and (ii) understand the role of morphometric indices in defining the vulnerabilities of lake ecosystems in Indian settings. We subsequently discuss our findings in order to find commonalities and disparities that will help to establish a methodological framework for further limnological studies. This study greatly improves our understanding of sediment depositional processes in the lake basins from a region of high biodiversity and conservation value. Further, this work provides useful insight for future management under the prism of climate change and other anthropogenic pressures.

2. Study area

Morphometric data from 35 lakes basins, covering a wide geographical terrain ranging from the Himalayan region in the north to peninsular India in the south were compiled (figure 1; table 1). The majority of the studied lake basins are characterised by their tectonic origin, and the remaining are categorised as glacial, fluvial or coastal origin. The Indian subcontinent exhibits highly variable climatic conditions in terms of temperature gradient and complex moisture pathways (Bookhagen and Burbank 2006; Owen *et al.* 2009; Wünnemann *et al.* 2010; Scherler *et al.* 2011). The precipitation over the Indian subcontinent is governed by the Indian summer monsoon (ISM) from June to September (JJAS), whereas westerlies and northeast (NE) monsoon contribute mostly in the extreme north of Himalaya and southern peninsula, respectively (Benn and Owen 1998; Bookhagen *et al.* 2005; Bhutiyani 2015). The elevation of studied lakes varies between ∼0 and ∼4500 m a.s.l. (meters above sea level), among which four are categorised as high-altitude lakes as they are located above 3000 m (Panigrahy *et al.* 2012). Furthermore, the surface area of the lakes ranges

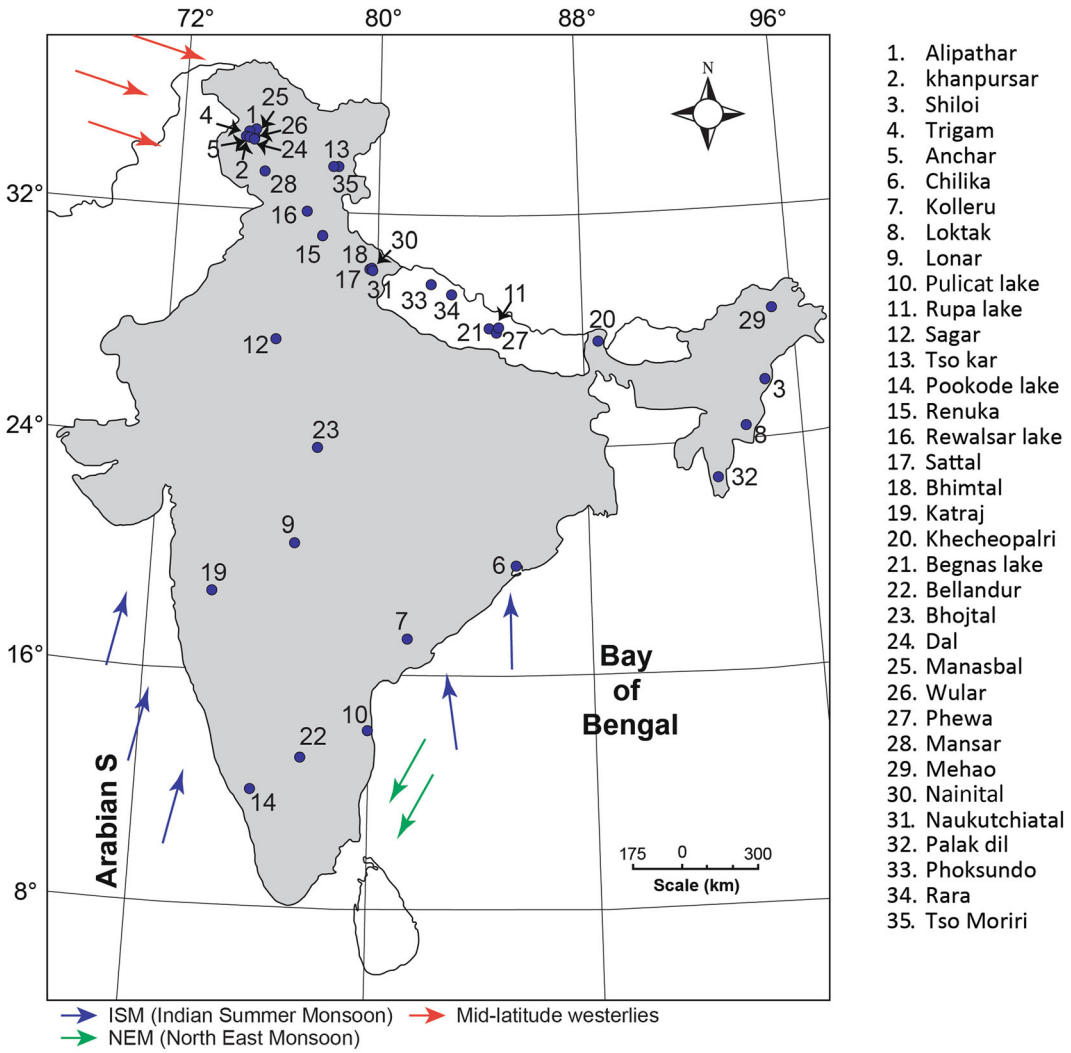


Figure 1. Indian map with major moisture regimes. Blue circles shown in the figure are the lakes discussed in the text.

between ∼0.026 and ∼1050 km^2, with the majority belonging to the medium-size category (>1 km^2 surface area) (Imboden 1990).

3. Methodology – data compilation and analysis

As discussed earlier, in terms of morphometric data availability, only a few works have been done in the Indian subcontinent. However, these works have mostly discussed the morphometric parameters in terms of fluvial systems (Bali *et al.* 2012; Kulkarni 2013; Dar *et al.* 2014). In the present study, the morphometric parameters were either compiled from the available sources (supplementary table S1) or were calculated following the methodology described in previously published literature (Håkanson 2005; Kolada *et al.* 2005; Stefanidis and Papastergiadou 2012). The geographical location of the lakes is given in decimal degrees, while the lake's altitude (in m a.s.l.) was either obtained from published literature or was retrieved from Google Earth Pro imagery (version 7.1.8.3036). Catchment and lake surface area, shoreline length, maximum and mean depths were obtained from published sources (supplementary table S1). When data were not available, we calculated them from Google Earth Pro imagery (version 7.1.8.3036). Furthermore, a linear regression equation between the maximum and mean water depth of the lake basin was used to estimate either of these two variables in case of missing values. The accuracy of the linear model and the reliability of data obtained from Google Earth Pro imagery was tested by estimating the normalised root mean square error (NRMSE) (supplementary table S2). The NRMSE values for mean and maximum depth calculated from the regression equation were 0.13 and 0.15,

Table 1. *Factor loadings for the morphometric parameters discussed in the text. Values with >0.50 loadings are shown in bold.*

	PC1	PC2
Size ratio	**0.75**	0.33
FL	0.43	**0.56**
CA	**0.84**	0.38
MxD	**–0.50**	**0.82**
MnD	**–0.51**	**0.83**
SA	**0.90**	0.39
SL	**0.77**	0.43
SR	0.35	**–0.50**
DR	**0.97**	–0.21
RD	**–0.97**	0.21
SDI	0.33	0.30
EF	**0.97**	–0.17

FL = Fetch length (m); CA = catchment area (km^2); MxD = maximum depth (m); MnD = mean depth (m); SA = lake surface area (km^2); SL = shoreline (km); SR = Schindler's ratio; DR = dynamic ratio; RD = relative depth; SDI = shoreline development index; EF = energy factor; LV = lake volume (km^3).

Size ratio = SA (km^2)/SL (km); SR = (CA + SA)/(LV) (Stefanidis and Papastergiadou 2012); DR = $(SA)^{1/2}$/MnD (Håkanson 2012); RD = $50\times MxD\times(\pi)^{1/2}/(SA)^{1/2}$ (Håkanson 2012); SDI = $(SL)/(2\times(\pi\times SA)^{1/2})$ (Lakewatch 2001); EF = $(SA)^{1/2}$/MxD (Håkanson 1981).

respectively, suggesting insignificant errors between predicted and observed values. Similarly, we have also calculated the uncertainties associated with the measurement of lake surface area (SA) and shoreline (SL) with google earth and observed values (based on literature). In this case, we have only selected the lakes which does not show much seasonal variation in the lake surface area. The lower NRMSE for lake surface area (0.13) and shoreline length (0.09) suggest the high reliability of google earth-based calculation in case of missing values.

The morphometric indices such as size ratio, Schindler's ratio (SR), dynamic ratio (DR), relative depth (RD) and energy factor (E) were also calculated following the methodology described in Håkanson (1976, 2005); Blais and Kalff (1995); Stefanidis and Papastergiadou (2012). The indices were evaluated using the following relations:

(a) Size ratio $= \frac{SA}{SL}$

(b) Schindler's ratio (SR) $= \frac{(CA+SA)}{LV}$

(c) Dynamic ratio (DR) $= \frac{\sqrt{SA}}{MnD}$

(d) Relative depth (RD) $= \frac{(50\times MxD\times\sqrt{\pi})}{\sqrt{SA}}$

(e) Shoreline development index (SDI) $= \frac{SL}{2\times\sqrt{\pi\times SA}}$

(f) Energy factor (EF) $= \frac{\sqrt{SA}}{MxD}$

where SA = lake surface area (km^2); SL = shoreline (km); CA = catchment area (km^2); LV = lake volume (m^3); MnD = mean depth (m); MxD = maximum depth (m).

Further, to facilitate the discussion of the results, the lakes were categorised into three groups based on the mean water depth, i.e., very shallow lake (water depth <3 m), shallow lake (3–15 m), and deep lake with a water depth of >15 m (Kolada *et al.* 2005), and in two groups based on the surface area, i.e., small lakes (<1 km^2) and large lakes (>1 km^2).

Principal component analysis (PCA) was performed on morphometric parameters using PAST software (version 3.13) in order to examine and identify the parameters which explain the highest share of variation in our data (Hammer *et al.* 2001). The PCA is a useful and widely used method that reduces the multivariate data into a meaningful and new set of *N*-principal components that are uncorrelated to each other (Håkanson 2005; Schäfer *et al.* 2014; Jollife and Cadima 2016; Prasad *et al.* 2020). Before PC analysis, Ln-transformation was performed to normalise the variable scales and highly skewed values of the 12 parameters i.e., FL, CA, MxD, MnD, LA, SL, SR, DR, RD, SDI, EF, and LV.

4. Results and discussion

4.1 *PCA of morphometric parameters*

The PC analysis extracted two principal components explaining ∼76% of the total variance (figure 2). The description of the components showed that PC1 (53% of the total variance) was significantly correlated with the size ratio and the lake surface area, suggesting that the first component constitutes a gradient of the size factor of the lake basin (figure 2a). Conversely, PC2 (∼23% variance) showed a significant positive correlation with lake water depth (figure 2b). Interestingly, the significant correlation between altitude and maximum water depth ($r = 0.52$; $N = 35$; $p < 0.05$) shows that the lakes at higher altitudes have greater water depth as compared to the lake basins situated at lower altitudes. In the high-altitude region, the sediment is mostly contributed by glacial melt or streams with relatively low

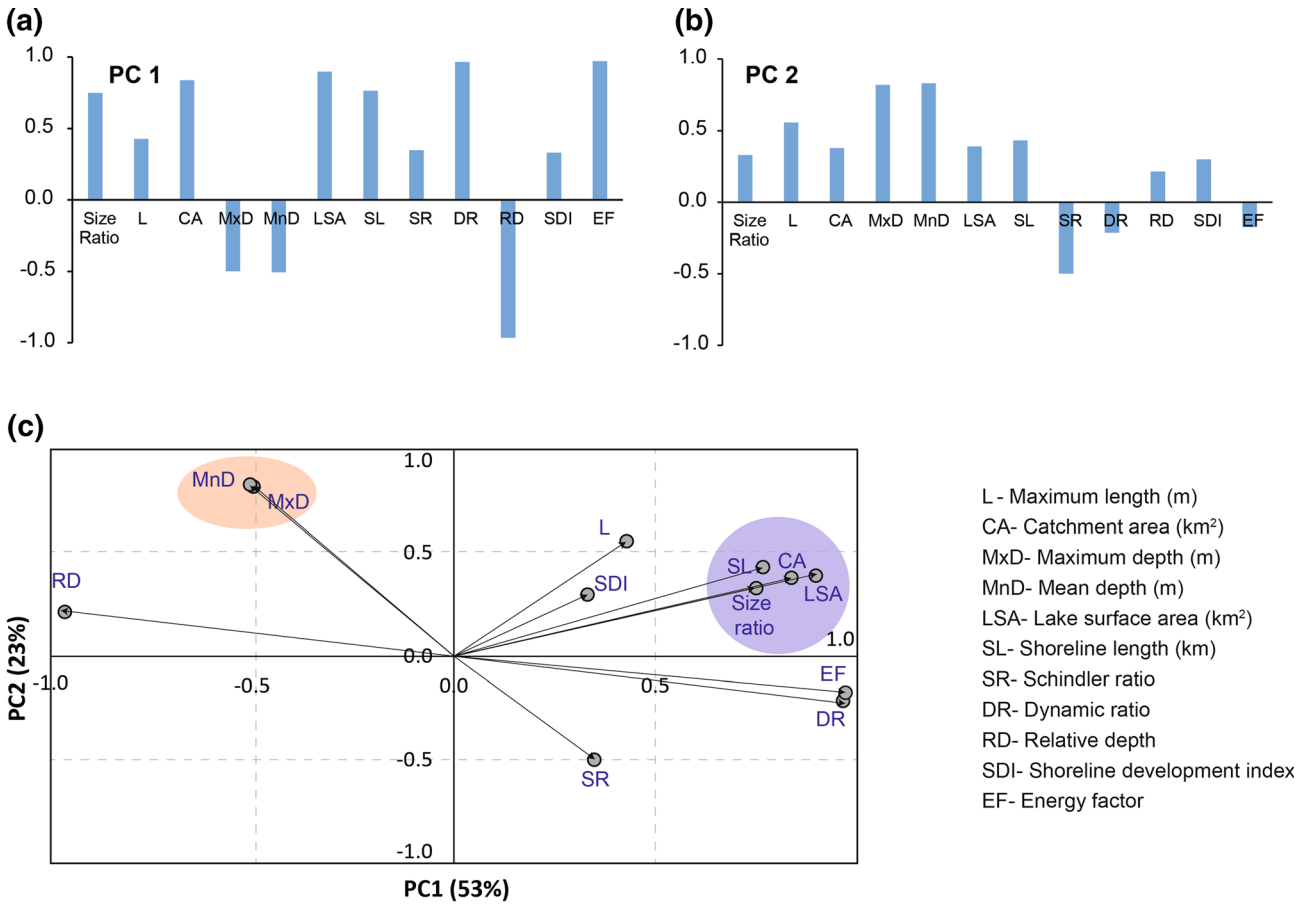


Figure 2. Principal component analysis of the morphometric parameters of the lakes. PC1 and PC2 loadings in (**a**) and (**b**), respectively; and (**c**) biplot between PC1 and PC2.

sedimentation rates. Although other factors, such as orographic controls, river discharge and catchment lithology, may also play a crucial role in controlling the sedimentation rate in a lake basin. Further, in low-altitude regions, the streams/inflow contribute a relatively huge amount of sediments which are consequently deposited in the basin. However, only based on the limited datasets, it is difficult to understand the impact of the lake's origin on the sediment supply. Therefore, detailed investigations or more datasets required from various geomorphological terrain to qualitatively assess the relation between water depth and the lake's origin.

4.2 *Lake basin morphometry*

4.2.1 *Schindler's ratio (SR)*

The SR [(catchment area + lake area)/lake volume] is commonly used to measure the combined effect of the catchment and the lake area on the lake water quality (Kolada *et al.* 2005; Stefanidis and Papastergiadou 2012). Higher SR values possibly indicate the strong influence of the catchment on the lake basin. As shown in figure 3(a) and supplementary table S1, the high SR values of Anchar, Tso Kar, Khecheopalri and Bellandur Lake suggest that the water quality is likely influenced by a much larger catchment area as compared to the lake water volume, and thus implies that the lake receives higher nutrition/sediments from the catchment. Similar findings have been observed in Greek lakes, where the lakes with higher SR were susceptible to high nutrient inputs, resulting in increased aquatic productivity (Stefanidis and Papastergiadou 2012). However, due to the lack of chlorophyll-*a* (a proxy used to indicate lacustrine productivity) data from most of the lake basins, it is beyond the scope of this study to establish the relation between SR and chlorophyll content or the aquatic primary productivity in the basin. Interestingly, the significant negative correlation ($r = -0.67$; $n = 16$; $p < 0.05$) between standardized SR with lake surface area for smaller lakes indicates that SR is more applicable to

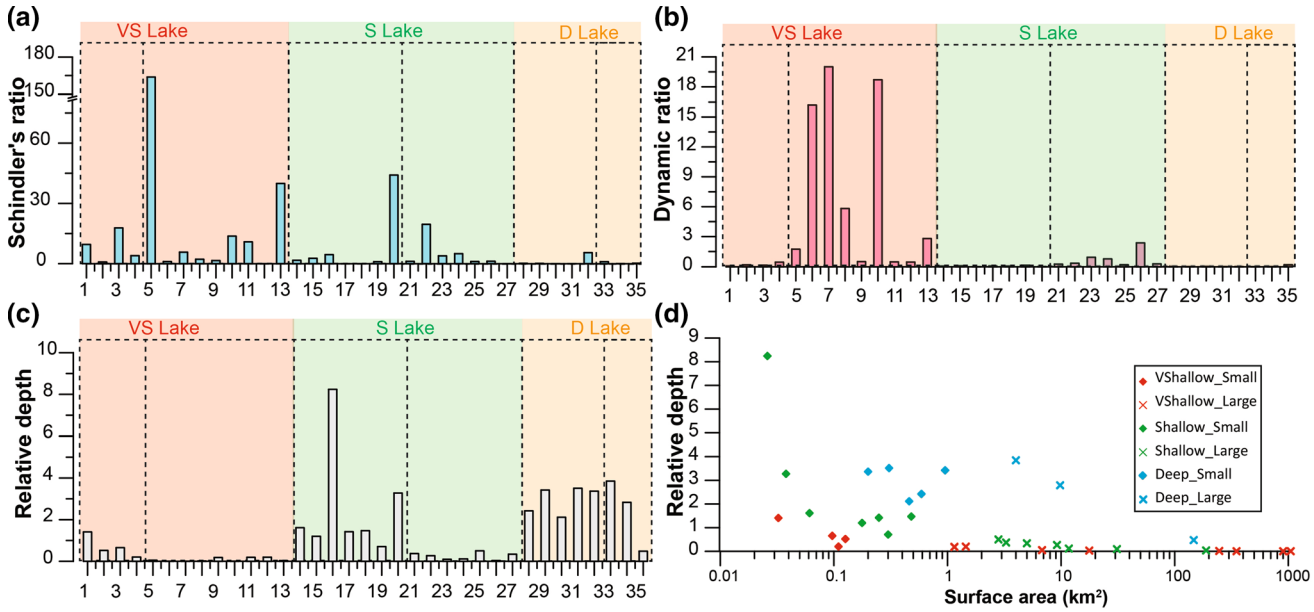


Figure 3. Plots of various morphometric parameters: (**a**) Schindler's ratio; (**b**) dynamic ratio; (**c**) relative depth; and (**d**) relative depth *vs.* surface area of the lakes. The lakes are grouped into three broad categories based on the maximum water depth (i.e., very shallow – VS, shallow – S and deep-water lake – D). The dotted lines further sub-categorised the datasets into two groups based on size of the lake.

understanding the catchment impact on the smaller lake basins. However, the larger lakes (>1 km^2 surface area) are less likely influenced by the catchment processes, probably due to larger water volume, which dilutes the catchment influence. In addition, the catchment slope and the presence of various geomorphic features such as terraces, drainage patterns and stream inflow may also influence the sediment contribution in the lake basin. For example, the presence of a steep slope in Lonar Lake (central India) catchment attributes to a higher sediment supply from the catchment, as evident from the higher concentration of lithogenic (e.g., Al and Ti) in the lake surface sediments (Basavaiah *et al.* 2014). Further, compared to the catchment relief, the catchment geology and anthropogenic activity may also play a crucial role in influencing sediment characteristics in the basin. Along with the organic matter, pollutant inputs and nutrients, the surface runoff contributes to sediments from the catchment. For example, in Dal (Kashmir), Mansar (Jammu), Renuka and Rewalsar Lake (Himachal Pradesh), the sediment characteristic is largely governed by the catchment geology and anthropogenic input into the lake basin. However, in the case of Tso Moriri (in NW Himalaya) and Lonar Lake (central India), the sediment characteristics of the lake basin are dominantly controlled by lake processes (such as evaporation) as compared to the catchment geology. This suggests that the interplay between the lake and the catchment processes along with the human activities in and around the lake basins, cumulatively controlled the sediment behaviour in the lake basins.

4.2.2 *Dynamic ratio (DR)*

The DR ratio is commonly used to understand the role of wave action in the lake basin (Håkanson 1981; Blais and Kalff 1995) (supplementary table S1). The DR of >0.8 points out that the distribution of the bottom sediment in the lake basin may also be controlled by the wave action (Håkanson 1981; Bachmann *et al.* 2000). Within the lake basin, the wave action is largely governed by wind direction, fetch length (the longest continuous distance of the lake), velocity and duration of the wind (Håkanson 1976, 1981; Smith 1991). However, the slope angle of the basin, water depth, size of the lake and human interventions in the basin are also considered as critical factors (Håkanson 1976, 1981). The bottom sediments in the lake basin with greater water depth and smaller fetch length are less influenced by wind-induced

mixing. The high DR values of the shallow-large lake systems such as Chilika, Kolleru, Loktak, Pulicat, Tso Kar and Wular Lake suggest that wave (wind) energy has an active role in controlling sediment distribution within the basin (figure 3b). However, in most of the lakes (e.g., Naukutchiatal, Khecheopalri, Bhimtal, Nainital, Phewa and Renuka Lake), the closed catchment (surrounded by dense vegetation/mountains) limits the role of wind-induced mixing. In addition, the relation between water depth and fetch length may also be considered as crucial factor to influence wind-induced mixing within the lake basin. The lower ratio between fetch length and maximum depth (FL/MxD) (as in Mansar, Rara, Alpather and Phoksundo) demonstrated that the influence of wind-induced mixing is limited due to the relatively greater water depth in the lake.

4.2.3 *Relative depth (RD)*

The relative depth indicates the susceptibility of the water column to mixing effects. Smaller values of RD reflect greater susceptibility to mixing effects (Stefanidis and Papastergiadou 2012) (supplementary table S1). Along with the SR, RD provides a preliminary overview of the water quality status (Nõges 2009; Stefanidis and Papastergiadou 2012) (figure 3c). The inverse relation between RD and the lake surface area indicates that the larger lakes are more susceptible to water mixing (figure 3d). Except for Phoksundo and Rara Lake, along with the longest fetch length, all the larger lake basins (>1 km^2) (e.g., Anchar, Chilika, Dal, Kolleru, Loktak, Tso Kar, Pulicat and Wular Lake) show extremely low RD values suggesting higher susceptibility to wind-induced mixing effects. The high DR values for the same lakes corroborate this result, pointing out the high likelihood of wave-induced mixing effects.

4.2.4 *Shoreline development index (SDI)*

The shoreline development index practically relates the shoreline length of a lake to the perimeter of a perfectly circular lake of equal area (supplementary table S1) (Lakewatch 2001). A value equal to 1 indicates a perfectly circular lake (Graf 1982; Blais and Kalff 1995; Håkanson 2005). Higher SDI values indicate complex shorelines that provide more habitats for aquatic biota, such as fish and birds (Graf 1982; Radomski and Goeman 2001; Oikonomou and Stefanidis 2020). In addition, the development of littoral vegetation is influenced not only by the available nutrients but also by habitat complexity (Kalff 2002; Ness 2006). The SDI values of studied lakes lie between 1 and 4.3, with a majority of lakes showing >1.5 SDI, suggesting irregular-shaped shorelines that promote complex littorals with increased habitat heterogeneity.

Along with other parameters (e.g., lake size, water depth, inflow, and outflow), SDI is a decisive factor in the sample collection from the basin. To capture the natural variability of sediment distribution within the lake, the SDI values can be utilised (Håkanson 1981). Higher SDI values suggest that a larger number of samples are required to represent the entire lake basin. Although this relationship is not the sole criterion to define the number of samples from the lake basins. Other criteria, such as the location of the samples (near shore or the deepest part of the basin), location near stream inputs and variable water depth, should also be considered as critical factors for the design and conduction of sediment sampling in a lake.

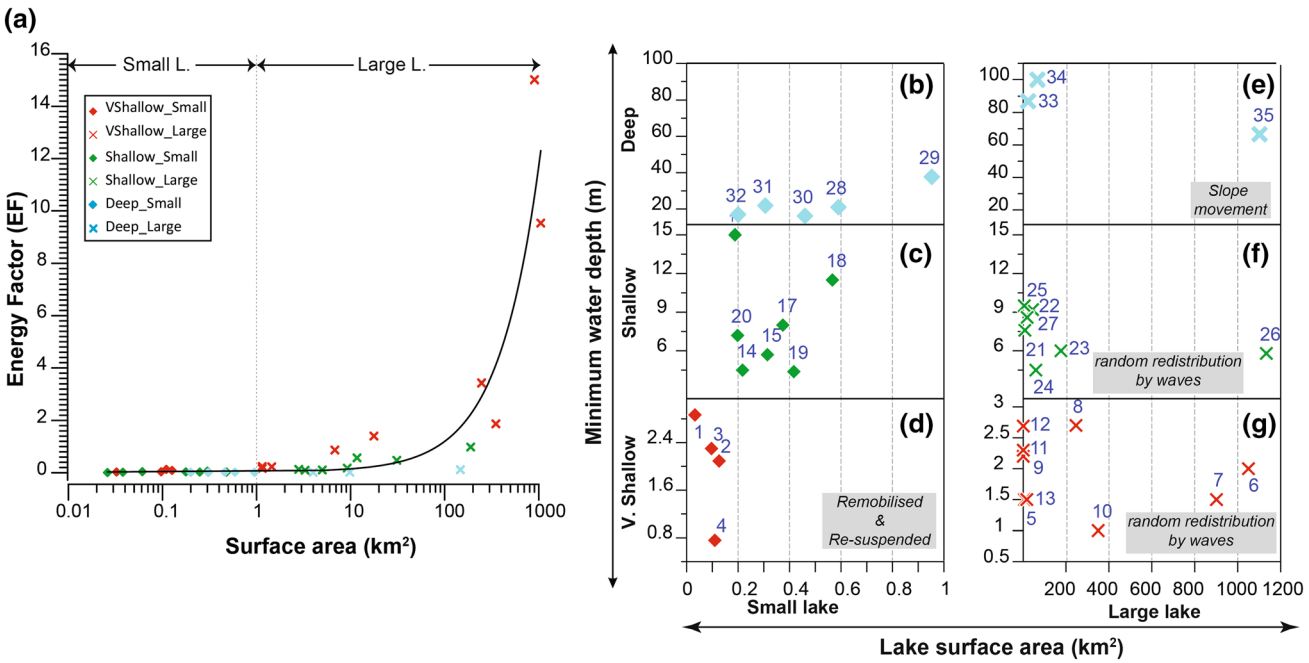
4.2.5 *Energy (E) factor*

The sediment distribution (accumulation + transportation) in the lake basin is the function of wave energy (E) which relays on the relation between the maximum fetch length, water depth, duration and direction of wind velocity, the topography of the region, and basin morphometry (Graf 1982). Therefore, it is important to investigate the relationship between these components in order to understand the dominant factors which control the sedimentation within the lake. The E-factor is calculated based on the relation between the surface area (which is the function of maximum fetch length) of the lake body and maximum water depth (supplementary table S1). In the lake basin, as the fetch length increases along with the wind velocity, the depth of wave penetration also increases and thus affects the sediment distribution and mixing of the water column (Bretschneider 1952; Li *et al.* 2017). However, in small lakes where the littoral zone is covered with aquatic vegetation, the role of wind-induced lake mixing is not significant. Further, in a smaller lake, the role of wave energy in controlling sediment distribution is limited (Håkanson 1981). The surrounding terrestrial

vegetation and the presence of aquatic vegetation in the shallow part of the lake reduce the effect of wind strength and hence the complete mixing of the lake water.

In our case, the higher E value for Kolleru, Chilika, Loktak, Pulicat and Tso Kar Lake implies that the larger wind fetch length and the surface area of the lake as compared to the very shallow water depth, supports wind-induced mixing (figure 4a). In other cases, we expect that the topography of the lake catchment may also play a key role in enhancing the wind-induced mixing of the water column. The shallow lakes from the Kashmir region (Alpather and Anchar) are situated between Pir Panjal and the Great Himalayan ranges (Rashid *et al.* 2013). Therefore, the increased wind strength due to the 'tunnelling effect' may also enhance wind velocity and thus the turbulent mixing, which affects the bottom sediments in the lake basin. Similarly, Tso Moriri Lake, which is situated between the high mountain ranges (∼4500 m a.s.l.), is also impacted by strong winds induced by similar mechanisms. However, due to the greater water depth (105 m), the role of wind-induced mixing of the bottom sediments is negligible (Mishra *et al.* 2014).

The relation between the surface area and maximum water depth manifests the dominant mechanisms (e.g., slope movement or slumping, wave activity, temperature gradient and peripheral wave activity) in controlling the sediment re-distribution within the lake basin (figure 4b–g) (Hilton 1985; Wu *et al.* 2019). In 1994, Larsen and Macdonald proposed a 'heuristic' approach to predict sediment mixing using surface area and maximum water depth in the lake basin. The bottom of the lakes with a small surface area but with larger maximum depth have minimal impact of the wind-induced mixing (figure 4b). However, in very shallow lakes with small surface areas (e.g., Alipathar, Khanpursar, Shiloi and Trigam Lakes), the bottom sediments are characterised by remobilised and re-suspended sediments (figure 4d). The sediment distribution in such smaller lakes is largely governed by peripheral wave activity and slumping due to the slope movement. Care should be taken while collecting the sediment samples from such a shallow zone in the lakes, as these regions are actively affected by wave action and bioturbations. Furthermore, in larger lakes (surface area >1 km^2) with relatively very shallow to shallow water depths (e.g., Anchar, Chilika, Kolleru, Loktak,



1. Alipathar; 2. Khanpursar; 3. Shiloi; 4. Trigam; 5. Anchar; 6. Chilika; 7. Kolleru; 8. Loktak; 9. Lonar; 10. Pulicat; 11. Rupa; 12. Sagar; 13. Tso Kar; 14. Pookode; 15. Renuka; 16. Rewalsar; 17. Sattal; 18. Bhimtal; 19. Katraj; 20. Khecheopalri; 21. Begnas; 22. Bellandur; 23. Bhojtal; 24. Dal; 25. Manasbal; 26. Wular; 27. Phewa; 28. Mansar; 29. Mehao; 30. Nainital; 31. Naukutchiatal; 32. Palak dil; 33. Phoksundo; 34. Rara; 35. Tso Moriri

Figure 4. The relation between various morphometric parameters: (**a**) a plot between energy factor (EF) and surface area of lakes; (**b–g**) scatter plots between maximum water depth and surface area. Here the plots are divided based on water depth (i.e., very shallow, shallow, and deep-water) and size of the lakes (small and large lakes).

Lonar, Pulicat, Rupa, Sagar, Tso Kar, Begnas, Bellandur, Bhojtal, Dal, Manasbal, Phewa and Wular lakes), the bottom sediments are largely governed by random redistribution by waves (figure 4f–g), whereas the bottom sediments in the lakes with both relatively larger surface area and greater water depth (e.g., Phoksundo, Rara and Tso Moriri Lake) are largely affected by the slope movement (especially in the shallow/near-shore region) in the lake basin (figure 4e).

4.3 *The implication of various morphometric indices*

The morphometric parameters are a crucial component which provides a primary approach to understanding the modern lake condition in terms of sediment distribution and the impact of the watershed on lake basins. These parameters are important to establish baseline information for further paleoclimate studies. In the case of the Indian subcontinent, only a little work has been done focusing on the modern monitoring of lake basins in terms of changes in lake morphometric parameters. This may result in over(under)-estimation of proxy response in terms of climate variability. Therefore, this review work based on the compilation of morphometric parameters from various lake basins from the Indian subcontinent provides a better understanding of present-day lake status. This may further help researchers to establish a modern understanding of the lake basin in terms of the lake's morphometric parameters and their role in sediment distribution at the lake bottom and catchment and the lake's interrelationship.

4.3.1 *Impact on lacustrine basins*

Several studies on the Indian lake systems have shown a considerable change in the past climatic condition over a region (Ghosh *et al.* 2014; Leipe *et al.* 2014; Mishra *et al.* 2015; Rawat *et al.* 2015; Sandeep *et al.* 2017; Bhushan *et al.* 2018; Ankit *et al.* 2021; Kumar *et al.* 2021). The paleolimnological studies largely involve the interaction between four components, i.e., morphometric parameters (e.g., lake dimensions, maximum depth and lake water volume), climatic factors (rainfall, temperature, wind), the catchment processes and anthropogenic activities (Håkanson 2005; Mishra *et al.* 2014; Bonk *et al.* 2015). This complex interaction has a major role in governing the sediment composition and aquatic productivity in the basin. Several earlier works have focused on the catchment processes and the lake's internal response to understand the climate variability in the region (Guilizzoni *et al.* 2006; Anoop *et al.* 2013; Mishra *et al.* 2014; Schillereff 2015; Li *et al.* 2017). For example, the larger catchment area, with several active inflows, carries a significant amount of sediment/nutrition into the lake basin. The stream inflow is dependent on two factors, i.e., natural forcings (climate, tectonic activity), as well as anthropogenic activities (Hansen *et al.* 2014). A slight change in any of these variables may affect the sediment/nutrition supply in the lake basin (Reddy and Char 2006; Najar and Khan 2012; Rashid *et al.* 2013). The population around these lakes is growing rapidly because of the opportunities for agriculture and tourism in the wider region. The fast-growing population is certain to impact the water quality and the aquatic life, leading to enhanced eutrophication and disturbance of the lake ecosystem (Saini *et al.* 2008; Rashid *et al.* 2013; Agnihotri *et al.* 2017; Lone *et al.* 2020). In addition, the population growth in a region as a result of rapid urbanisation in the lake's watershed modifies the catchment characteristics in terms of (a) change in the slope of the stream channel; (b) sediment contribution; and (c) soil cover. The disturbance in any of the parameters in the catchment impacts the water quality and the aquatic life, leading disturbed lake ecosystems (Saini *et al.* 2008; Rashid *et al.* 2013; Agnihotri *et al.* 2017; Lone *et al.* 2020). The rapid growth of urbanisation because of the opportunities for agriculture and tourism in the Himalayan Lakes (e.g., Dal, Manasbal, Wular, Anchar, Nainital, Loktak) has hugely impacted these aquatic systems and their ecosystems.

The principle of paleoclimate studies using the lacustrine sediments lies in three basic criteria, (i) minimal mixing of sediments, (ii) sediment core should be chronologically constrained, and (iii) location of the core should be representative of the lake basin (Larsen and Macdonald 1994). Sediment mixing in the lake basin is a common phenomenon that occurs when re-suspended, remobilised sediments are deposited at the lake bottom by physical (wind, current, gravity) or biological (organic) activity (Larsen and Macdonald 1994). The wave action has maximum energy at the shore, which increases the redistribution and remobilisation of sediments. Therefore, the sediments near shore are

often disturbed due to wind-induced mixing. Thus, the sediment sampling at the shore focusing on understanding the spatial distribution of physical or chemical characteristics of sediment distribution in the basin is not recommended. However, the wave energy goes down towards the centre or maximum depth in the basin (Larsen and Macdonald 1994). Therefore, for paleoclimate studies, sediment sampling from the deeper part of the basin is crucial, as these regions are minimally impacted by the peripheral wave action, thus reducing the chronological uncertainties.

4.3.2 *Degradation state of lake basins*

Lakes are of great importance for the present society because of their domestic as well as industrial utilisation (Bronmark and Hansoon 1998). Apart from that, the lacustrine systems are the prominent storehouse to understand the past climate variability together with its direct link to the increasing anthropogenic activities in the surrounding region (Adrian *et al.* 2010). Out of the total contributions of freshwater, ∼22% is contained by the groundwater and ∼0.3% includes lakes and rivers, which are rigorously used for domestic, industrial and agricultural purposes (source: https://www.ngwa.org/). The understanding of the rapid degradation of lake systems due to the impact of natural as well as anthropogenic stressors is vital to improve our knowledge of lake restoration and management (Bronmark and Hansoon 1998; Bachmann *et al.* 2000; Vincent 2009). Therefore, the morphometric studies of the lake basin can provide an understanding of the factors influencing the vulnerability of the lake ecosystems. The lake degradation is dependent on multidirectional changes in and around the lake basins, e.g., (a) eutrophication – in response to change in nutrient supply into the lake basin, (b) change in lake volume and water depth, (c) variability in catchment processes (e.g., sediment supply through surface runoff); and (d) human activity (such as field and agricultural practices, water withdrawal due to irrigation). As suggested earlier, two factors, i.e., lake water depth and size of the lake, are the crucial components to affect the lake's ecosystem in terms of sediment distribution and degradation status. The changes in the lake area, which consequently affect lake basin morphometry may be caused due to anthropogenic as well as natural factors. Human activity has a significant impact on the lake basin. This effect is even more pronounced in lakes with huge catchment areas and shallow water depths, such as Anchar, Katraj, Khecheopalri, Palak Dil and Phoksundo Lakes. In addition, the low or decreasing rainfall in the lake's watershed may also affect the lake morphometry in terms of changes in the local hydrological conditions and the contribution of sediments from surface runoff. Usually, during increased rainfall conditions, the surface runoff contributes a huge amount of sediments to the lake bottom. In such cases, because of the high siltation, the shallow lakes are most vulnerable in terms of their ecosystems. In the present-day scenario, in the Indian subcontinent, several lakes from the Himalayas (Wular, Manasbal, Nainital, Dal and Loktak Lakes), peninsular India (e.g., Kolleru Lake, Lonar, Pookode and Sagar Lakes) and other coastal lakes (such as Pulicat, Chilka) are at the highest risk of shallowing due to increasing siltation and dumping of various dissolved solids into the lake basin. However, due to the scarcity of datasets in terms of seasonal change in the lake level due to these factors, it is highly difficult to decouple the causal mechanisms of deterioration of the lake basins.

Further, the inter-relationship between the lake's area and watershed may also be used to understand the lake's degradation status. For example, the higher Schindler ratio (SR) is an indicator of excess nutrient input from the catchment region, which implies higher productivity and increases degradation in the lake leading to the formation of dead zones in the water bodies (Stefanidis and Papastergiadou 2012; Kolada *et al.* 2017). This indicator is well established for shallow lake basins as in Anchar, Begnas, Khecheopalri and Tso Kar lakes. The extent of degradation is even more critical for the small and shallowest lake basin which is susceptible to enhanced productivity. For example, the lake ecosystems from a few of the selected shallow-small lake basins (e.g., Khanpursar and Trigam lakes from the Kashmir region) are degrading rapidly due to increased anthropogenic activity and excessive nutrition supply from the catchment resulting in rapid deterioration of the lake's ecosystem. The higher input of nutrients and increasing urbanisation in Anchar Lake catchment trigger the deterioration of the lake ecosystem. Additionally, the highly suspended material in the form of sediment load and dissolved solids is also an indicator of a threat to the lakes and their life forms. However, in larger lakes (with

low SR), even though the lake catchment is highly affected by anthropogenic activity due to its large size, and deeper water depth, their impact is comparatively minimal.

The shoreline development index is an important criterion to define the presence of aquatic biota and their habitat. However, due to increasing anthropogenic activity, these habitats are declining. Therefore, shoreline management is crucial to prevent excessive anthropogenic activity and degradation of the lake's shoreline habitats. In a review work by Reddy and Char (2006), several steps have been proposed to restrict anthropogenic activity around the lakes. They have mainly emphasised (i) restriction of building construction near the shorelines; (b) construction of community toilets to prevent human wastes into the basin; and (c) demarcation of shorelines by fences, roads, etc., which can aid in better conservation of these eminent water resources in the Indian subcontinent.

5. Conclusions

In this study, several morphometric parameters of 35 lakes from the Indian subcontinent and their role in controlling the sediment distribution in the basin have been studied. By comparing the morphometric properties of various lake basins, we have identified that the maximum water depth and the size of the lakes are crucial components to control the sediment distribution in the basin. The study provides preliminary information to establish the link between catchment and lake processes taking into consideration the climate variability in the region. In addition, we identified several morphometric indices (e.g., Schindler's ratio: SR, dynamic ratio: DR, relative depth: RD, shoreline development index: SDI and energy factors: EF) to investigate their role in influencing the degradation status of the lake ecosystem. The Schindler's ratio provides an understanding of the impact of the catchment process on the lake basin. The higher SR values for Anchar, Tso Kar, Khecheopalri and Bellandur lakes suggested that the lake degradation status in these lake basins is principally influenced by the larger catchment area. However, in larger lakes, the impact of the catchment is minimal due to the dilution by greater lake volume. Further, the relative depth (RD) provides an understanding of the susceptibility of the water column to mixing effects. The inverse relation between RD and lake surface area suggested that the larger lake basins (e.g., Anchar, Chilika, Dal, Kolleru, Loktak, Tso Kar, Pulicat and Wular lakes) are susceptible to water mixing due to wind-induced mixing. The water mixing due to the wave action has maximum energy at the shore resulting in redistribution and remobilisation of sediments. Therefore, in a study focusing on understanding the spatial distribution of physical or chemical characteristics of sediment distribution in the basin, we recommend focusing on the samples from the deeper part of the basin to avoid any wave action-induced sediment mixing. Furthermore, among the studied lakes, the majority of lakes have SDI > 1.5, suggesting irregularly shaped shorelines that promote complex littorals with increased habitat heterogeneity. Several lake basins, such as Begnas, Loktak, Wular, Renuka, and Sattal, with SDI > 1.5 suggest hotspots for higher sedimentary deposition either due to anthropogenic activity or increased input from the surface runoff.

Overall, this study suggested that for lake management identifying the morphometric parameters is crucial. In addition, the study provides baseline information, which can help to prevent and control the deterioration of the lake's ecosystems. However, a detailed understanding of the increased anthropogenic activity, natural climate variability, and catchment geology is crucial in order to comprehend the impact of these variables on lake ecosystems.

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Author statement

SS and PKM: Designed the research, interpreted the data and wrote the manuscript. KS: provided critical review and comments on the draft and technical expertise on data analysis.

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